

REWEAR-FUSED Backend Technical Specification

Cox Enterprises "Play With Purpose" — Carter's "Making & Remaking" Track | 72-hour build | Demo June 17, 2026

TL;DR

- In 72 hours, REWEAR-FUSED's backend can credibly precompute (not run live) a state-of-the-art three-engine co-design pipeline: Engine B uses RFdiffusion2 + LigandMPNN + Boltz-2 + PLACER + FoldSeek to generate a de novo Ser-His-Asp carbamate hydrolase scaffolded around a UMG-SP2-anchored transition state; Engine A uses RDKit-driven combinatorial enumeration with polyBERT/PolyID/PI1M-trained surrogates to constraint-filter $\sim 10^4$ aliphatic PU-urea elastane candidates; Engine C is a LightGBM multi-label compliance classifier wrapped in a FastAPI service emitting an ESPR-aligned Digital Product Passport JSON.
 - The grand-prize centerpiece is Engine B's RFdiffusion2 → LigandMPNN → Boltz-2 → PLACER pipeline — the exact stack used in Lauko et al. *Science* 2025 (de novo serine hydrolases) and Kim/Woodbury/Ahern *Nature* 2026 (de novo zinc metallohydrolases; published online Dec 3, 2025, in print as *Nature* 649:246–253, kcat/KM up to 53,000 M⁻¹ s⁻¹ and kcat up to 1.5 s⁻¹ from only 192 designs across two rounds). RFdiffusion2 (GitHub public release Sept 15, 2025; Ahern et al., *Nat. Methods* 23:96–105, Jan 2026, DOI 10.1038/s41592-025-02975-x) directly scaffolds atom-level theozymes including Ser-His-Asp triads — the exact differentiator versus the wild-type UMG-SP2 Ser-Ser-cis-Lys amidase-signature triad (Świderek et al., *JACS* 2025, DOI 10.1021/jacs.5c13147).
(Nature + 4)
 - The honest 72-hour reality: the backend ships *artifacts* (PDB + JSON), not a live GPU pipeline. All RFdiffusion2/Boltz-2 jobs are pre-baked on RunPod H100s (H100 PCIe spot \$1.99/hr, H100 SXM spot \$2.69/hr per RunPod 2026 pricing) across a ~48-hour compute window for ~\$400–500 total. The frontend reads static artifacts. Wet-lab validation is impossible in 72 h; the spec is explicit about what is *demonstrably credible in silico* versus the forward-looking vision. (Deploybase) (Deploybase)
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Key Findings

1. **RFdiffusion2 is the correct backbone-generation tool**, not RFdiffusion or RFdiffusionAA. It is the only open-source generative model that scaffolds *atom-level* theozymes (Ser, His, Asp functional groups around a transition state) without inverse-rotamer enumeration. It

solved 41/41 active sites on the AME benchmark versus 16/41 for prior methods, and active catalysts were found by testing fewer than 96 designs each across three reaction classes (Ahern et al., *Nature Methods* 23:96–105, January 2026, DOI 10.1038/s41592-025-02975-x; GitHub: RosettaCommons/RFdiffusion2). (PubMed Central + 2)

2. **LigandMPNN, not vanilla ProteinMPNN, is mandatory** because the carbamate transition state must be modeled atomically in the active site; LigandMPNN's native-backbone sequence recovery on small-molecule-contacting residues is 63.3% vs ProteinMPNN's 50.5% (Dauparas et al., *Nat. Methods* 2025, DOI 10.1038/s41592-025-02626-1). (Bakerlab)
3. **Boltz-2 (MIT license, released June 6, 2025) is the correct structure-prediction filter** — co-folds protein+ligand, predicts binding affinity in ~20 s on a single GPU, *commercially usable*, unlike AlphaFold3 weights (non-commercial) and ESM3 (restricted). Chai-1 (Apache 2.0) is a viable fallback. (Pubs)
4. **PLACER is the critical, paper-vetted active-site filter** (Lauko et al., *Science* 388:eadu2454, 2025; GitHub: baker-laboratory/PLACER) — predicts an ensemble of active-site conformations across the reaction coordinate, used to discard designs lacking preorganization. (PubMed Central + 2)
5. **Engine A cannot use polyBART/polyBERT/POLYT5 directly on segmented block copolymers** — Savit et al. (arXiv:2506.04233) explicitly state polyBART "is currently limited to homopolymers and does not yet support the representation or generation of copolymers, polymer blends, or polymer composites/formulations." Strategy: **representation decomposition** — predict each block's properties via polyBERT/PolyID, recompose via Fox/rule-of-mixtures, then fine-tune mechanical properties on the Ding et al. PU-elastomer XGBoost/RF/SVR/ANN/kNN ensemble (*Chin. J. Polym. Sci.* 41:422–431, 2023, DOI 10.1007/s10118-022-2838-6; R^2 0.85–0.91 on the benchmark dataset for Young's modulus, tensile strength, elongation at break). (arXiv + 2)
6. **Engine C's regulatory backbone is real and queryable**: CPSC Recalls REST API at (saferproducts.gov/RestWebServices/Recall?format=json), EU Safety Gate RAPEX weekly Excel dumps, plus statutory text for CPSIA, REACH Annex XVII, California AB 1817 (100 ppm Total Organic Fluorine from Jan 1, 2025; 50 ppm from Jan 1, 2027), New York ECL §37-0121, OEKO-TEX Standard 100 Class I, ESPR Regulation (EU) 2024/1781 with EU Central DPP Registry going live July 19, 2026. (LegiScan) (Fiegenbaum Solutions)

PART 1 — ENGINE B: DE NOVO CARBAMATE HYDROLASE PIPELINE

This is the grand-prize centerpiece. The pipeline mirrors Lauko et al. *Science* 2025 and Kim/Woodbury/Ahern *Nature* 2026 — both papers publish their pipeline code as Baker-lab GitHub repos. The novelty is the **target reaction**: hydrolysis of an aliphatic carbamate in a PU-urea soft segment by a designed Ser-His-Asp triad at near-neutral pH, anchored on (not copied from) the UMG-SP2 transition state.

1(a) Backbone generation: RFdiffusion vs RFdiffusion2 vs RFdiffusionAA

Tool	Repo	License
RFdiffusion (v1)	github.com/RosettaCommons/RFdiffusion	BSD-style
RFdiffusionAA	github.com/baker-laboratory/rf_diffusion_all_atom	non-commercial research; weights at files.ipd.uw.edu/pub/RF-All-Atom/weights/RFDiffusionAA_p GitHub
RFdiffusion2	github.com/RosettaCommons/RFdiffusion2	research/academic (apptainer SIF; nc

RFdiffusion2 specifics (Ahern et al., *Nat. Methods* 23:96–105, January 2026, DOI 10.1038/s41592-025-02975-x; bioRxiv 2025.04.09.648075 posted April 10, 2025; GitHub public release Sept 15, 2025): [Nature](#) [bioRxiv](#)

- **41/41 active sites** scaffolded on the AME (Atomic Motif Enzyme) benchmark **versus 16/41** for prior state-of-the-art. [bioRxiv](#)

- For three distinct catalytic reactions, active catalysts were identified by experimentally testing **fewer than 96 designs each** (Lauko-style funnel scale REWEAR-FUSED reproduces in silico). (bioRxiv)
- Uses flow-matching training with atomic-level conditioning; outputs protein backbones positioning user-specified atomic functional groups (Ser-O γ , His-N ϵ , Asp-O δ) within tight geometric tolerance. (Uw)
- Container image via apptainer/Singularity SIF; NVIDIA GPU required.

1(b) Sequence design: ProteinMPNN vs LigandMPNN vs SolubleMPNN

Use **LigandMPNN** — Dauparas et al., *Nat. Methods* 2025, DOI 10.1038/s41592-025-02626-1; repo (github.com/dauparas/LigandMPNN), **MIT license**; model weights at (files.ipd.uw.edu/pub/ligandmpnn/). (GitHub)

- ProteinMPNN (Dauparas et al., *Science* 2022) **cannot see the substrate** — protein backbone atoms only.
- LigandMPNN explicitly models all nonprotein atoms (ligand, metals, nucleotides). Native sequence recovery on residues within 5 Å of a small molecule: **63.3% (LigandMPNN) vs 50.4% (Rosetta) vs 50.5% (ProteinMPNN)**; **50.5% vs 35.2%/34.0%** on nucleotides; **77.5% vs 36.0%/40.6%** on metals. (Bakerlab)
- LigandMPNN also generates side-chain conformations (χ_1 - χ_4), enabling pre-folding evaluation of catalytic geometry. (Nature)
- SolubleMPNN: an alternative ProteinMPNN weight set biased away from hydrophobic surface residues — used as a *post hoc* re-design pass on solvent-exposed positions; not the primary sequence designer.

Sampling: T = 0.1 for catalytic-pocket residues (use `--fixed_residues` and `--bias_AA` to lock Ser-His-Asp identities/positions from RFdiffusion2); T = 0.2-0.3 elsewhere. Generate **8 sequences per backbone**.

1(c) Structure prediction / validation

Tool	License	Commercial?	Strength	Verdict
AlphaFold2 / ColabFold	Apache 2.0	Yes	Fast single-chain via MMseqs2 MSA server	Sequence-only self-consistency check
AlphaFold3	DeepMind terms; weights	No	Best protein+ligand co-folding accuracy	Excluded by license

Tool	License	Commercial?	Strength	Verdict
non-commercial				
ESMFold / ESM3	ESM3 weights non-commercial; ESMFold academic	ESM3 No	Sequence-only ultra-fast	Backup speed pass; can't model carbamate
Boltz-1	MIT	Yes	AlphaFold3-level co-folding (Mit)	Acceptable
Boltz-2	MIT (released June 6, 2025)	Yes	Co-folds protein+ligand AND predicts affinity at FEP-comparable accuracy in ~20 s/GPU; experimental method conditioning; distance/pocket constraints (bioRxiv)	PRIMARY co-fold filter
Chai-1	Apache 2.0	Yes	AlphaFold3-comparable, MSA-optional (Medium)	Solid cross-check

Filter thresholds (matched to Lauko 2025 / Kim 2026 funnels):

- **AF2/Boltz-2 self-consistency RMSD** (designed backbone vs predicted C α): **< 2.0 Å full-chain; < 1.0 Å for catalytic-triad residues.**
- **pLDDT**: mean > 80 across protein; > 90 at active-site residues.
- **PAE**: < 5 Å between any pair of catalytic-triad residues.
- **Boltz-2 (affinity_probability_binary)**: > 0.5 against the carbamate (binder/decoy classifier — NOT an absolute K_d). (GitHub)
- **Boltz-2 confidence score** (= 0.8·pLDDT + 0.2·ipTM per the Boltz-2 documentation): > 0.85. (ResearchGate)

1(d) Active-site / catalytic geometry validation: PLACER

PLACER (Protein-Ligand Atomistic Conformational Ensemble Reproduction/Resolver) — Lauko et al. *Science* 2025 (science.adu2454); repo (github.com/baker-laboratory/PLACER); bundled in (github.com/laukoag/serine-hydrolase-design) (recommended fork target).

- **Inputs**: protein + ligand at each catalytically relevant state (Michaelis complex, tetrahedral intermediate, acyl-enzyme, second tetrahedral intermediate).
- **Outputs**: ensemble of active-site conformations per state.

- **Reported metrics:** catalytic-residue RMSD across the ensemble (preorganization); oxyanion-hole H-bond occupancy to the carbonyl O of the tetrahedral intermediate; Ser-Oy → carbonyl-C distance (target median < 3.0 Å, low variance); His-Nε → Ser-Oy distance (< 3.5 Å); Asp-Oδ → His-Nδ distance (< 3.0 Å).
- **Why this is the breakthrough:** Lauko 2025 showed that two oxyanion-hole H-bond contacts (vs one) shift the catalytic Ser preferentially to the catalytically-productive g⁻ rotamer; PLACER ensemble analysis across reaction states is what made their best hydrolases >60,000× more efficient than prior cysteine-based de novo enzymes. (Bakerlab)

1(e) Catalytic theory: Ser-His-Asp at neutral pH vs UMG-SP2 Ser-Ser-cis-Lys at alkaline

Natural UMG-SP2 — discovered by Branson et al., *Angew. Chem. Int. Ed.* 62:e202216220, 2023 (DOI 10.1002/anie.202216220); QM/MM dissected by Świderek et al., *JACS* 147:42511–42523, 2025 (DOI 10.1021/jacs.5c13147); independent QM/MM by Paiva et al., *Chem. Sci.* 16:2437–2452, 2025 (DOI 10.1039/D4SC06688J); crystallographic engineering by Li et al., *Adv. Sci.*, 2025, DOI 10.1002/advs.202416019:

- Metagenome-derived urethanase; hydrolyzes >90% of **TDA-diethylene-glycol dicarbamate (20 mM) in 24 h** (Branson 2023). (ResearchGate)
- Belongs to the **amidase-signature family** with a **Ser-Ser-cis-Lys** triad. (ChemRxiv)
- QM/MM (Świderek 2025) reveals a 3-step esterase-like mechanism: (1) acylation by the nucleophilic Ser; (2) hydrolysis (deacylation) by catalytic water; (3) decarboxylation of carbamic acid leaving group → amine + CO₂. Rate-determining-step ΔG[‡] = **21.2 kcal/mol** (calculated) vs **22.9 kcal/mol** (experimental TOF). (Cambridge)
- Constant-pH neMD/MC simulations show the active-site Lys is deprotonated (catalytically competent) only at **alkaline pH** — UMG-SP2 needs basic conditions to function. (nih)

REWEAR-FUSED's choice — classical Ser-His-Asp at near-neutral pH:

- His pK_a ~6–7 → activity at pH 6–8, far cheaper and safer for a children's-apparel recycling process than alkaline conditions.
- Lauko et al., *Science* 388:eadu2454, 2025 (DOI 10.1126/science.adu2454) demonstrated **de novo design of exactly this triad for esters**, reporting **k_{cat}/K_M up to 2.2 × 10⁵ M⁻¹ s⁻¹** and **k_{cat} = 0.076 s⁻¹** (pH 7.0, 25 °C), "yielding an estimated rate enhancement of over 10⁸." Engine B is the carbamate analog of this design. (Bakerlab)
- Mechanism: Asp polarizes His → His-Nε general-base abstracts Ser-Oy proton → Ser-Oy attacks carbamate C → tetrahedral intermediate stabilized by oxyanion hole (two backbone-NH donors) → collapse to acyl-enzyme + amine leaving group → water-mediated deacylation → carbamic acid → spontaneous decarboxylation → free polyol + CO₂.

- Kim/Woodbury/Ahern, *Nature* 649:246–253, 2026 (published online December 3, 2025; DOI 10.1038/s41586-025-09746-w) on de novo zinc metallohydrolases — verbatim abstract: "From an initial set of 96 designs tested experimentally, the most active has a catalytic efficiency (kcat/KM) of 16,000 M⁻¹ s⁻¹... A second round of 96 designs yielded 3 additional highly active enzymes, with kcat/KM values of up to 53,000 M⁻¹ s⁻¹ and a catalytic rate constant (kcat) of up to 1.5 s⁻¹." — orders of magnitude higher than previously designed metallohydrolases. [PubMed Central](#) [Nature](#)

Engineering anchor (not a copy): REWEAR-FUSED's theozyme inherits *substrate orientation and oxyanion-hole geometry* from the UMG-SP2 ligand-bound co-structures (Li et al., *Adv. Sci.* 2025, at 2.16 Å and 2.40 Å resolution; hydrophobic-pocket residues L319/L323/W143/W382 that bind the alkyl carbamate). The *catalytic residue identities* are swapped to Ser-His-Asp before RFdiffusion2 scaffolding. [nih](#) [Wiley Online Library](#)

1(f) Docking / substrate modeling

1. Carbamate SMILES → 3D conformer via RDKit ([AllChem.EmbedMolecule](#)) + [UFFOptimizeMolecule](#).
2. RFdiffusion2 encodes substrate atoms (carbonyl C, carbonyl O, leaving-group N) as atom-level conditioning alongside the Ser/His/Asp triad.
3. LigandMPNN sequences the pocket with substrate present.
4. Boltz-2 co-folds the designed sequence with the carbamate SMILES (YAML config with [protein](#) and [ligand](#) blocks).
5. **GNINA** ([github.com/gnina/gnina](#)) and/or **AutoDock Vina** as orthogonal docking cross-checks (binding-energy estimate only; Boltz-2 pose is authoritative).
6. PLACER runs on each reaction-coordinate state (Michaelis, TS1, acyl-enzyme, TS2) for preorganization metrics.

1(g) Compute reality and design funnel

Lauko 2025 / Kim 2026 funnel scale: thousands of RFdiffusion2 backbones → tens of thousands of LigandMPNN sequences → AF2/Boltz-2 self-consistency → PLACER preorganization filter → 96 designs ordered for experimental testing. REWEAR-FUSED reproduces this funnel *in silico* and ships the top ~10.

Realistic 48-h compute budget on 4× NVIDIA H100 80 GB (RunPod or Lambda):

Stage	Per-design cost	Target count	Wall time on 4× H100
RFdiffusion2 backbones	30–90 s	2,000	4–8 h
LigandMPNN (8 per backbone)	5 s	16,000	6 h
Boltz-2 co-fold	20–40 s	~8,000 (top half)	12–18 h
PLACER (4 reaction states)	10 s/state	top 500	2 h
FoldSeek novelty	seconds	top 50	<1 h
Total			~30–40 h (fits 48 h)

Total spend: ~\$2.50–3.00/H100-hr × 4 × 40 h = **\$400–500**. **RunPod Community Cloud H100 PCIe at \$1.99/hr spot and H100 SXM at \$2.69/hr spot** (RunPod 2026 pricing). **Lambda Labs H100 80GB at \$2.99/hr on-demand** (April 2026); H100 SXM in 8-GPU configs at \$3.99/GPU/hr. **Vast.ai** as fallback at \$0.90–4.10/hr depending on host quality. (Deploybase + 4)

1(h) FoldSeek for novelty

FoldSeek — van Kempen et al., *Nature Biotechnology* 42:243–246, 2024 (DOI 10.1038/s41587-023-01773-0); repo (github.com/steineggerlab/foldseek); web server (search.foldseek.com):

- Submit each top design's predicted PDB against **AlphaFoldDB v4** and **PDB100**.
(UCSF Chimera)
- **TM-score < 0.5 against the closest hit** = "different fold" — the threshold Lauko 2025 used to argue their de novo serine hydrolases occupy folds *distinct from natural serine hydrolases*.
- ~6 seconds per query for AlphaFoldDB v1 (paper benchmark, vs 33 h for TM-align and 10 days for Dali on the same task); all top-50 designs FoldSeek'd in minutes. (Nature)
- Report: closest natural homolog's PDB ID, TM-score, sequence identity. Grand-prize claim: "structurally novel fold designed from scratch to act on a non-natural plastic substrate."

PART 2 — ENGINE A: POLYMER CONSTRAINT-GUIDED VIRTUAL SCREENING

Locked design space: aliphatic diisocyanate (HDI, H12MDI, IPDI — never aromatic MDI/TDI to avoid MDA/TDA carcinogens) × soft segment (PCL-diol Mn≈2000 g/mol or polycarbonate diol

Mn≈2000 g/mol; cleavable bond lives here) × chain extender (short diols/diamines) × hard-segment 25–35 wt%.

2(a) Polymer representation

Notation	Handles	Source	REWEAR-FUSED use
PSMILES	Homopolymers; [*] chain ends arXiv	Kuenneth & Ramprasad, <i>Nat. Commun.</i> 14:4099, 2023, DOI 10.1038/s41467-023-39868-6	Each block independently
BigSMILES	Hierarchical block/stochastic copolymers	Lin et al., <i>ACS Cent. Sci.</i> 2019	Full PU-urea segmented chain in artifact JSON
SELFIES/PSELFIES	Robust string for generative models	polyBART, arXiv:2506.04233, May 2025	Optional
Plain SMILES	RDKit descriptors	Landrum, RDKit	Chain-extender + isocyanate enumeration

Why segmented PU-urea is intrinsically hard: polyBART (Savit et al., arXiv:2506.04233, May 2025) explicitly states it "is currently limited to homopolymers and does not yet support the representation or generation of copolymers, polymer blends, or polymer composites/formulations." polyBERT and POLYT5 share this limitation. **Engine A therefore cannot use a single end-to-end polymer language model — it must decompose-predict-recompose.**

2(b) Property prediction models

Model	Repo	License	Properties	REWEAR-FUSED rol
polyBERT	github.com/Ramprasad-Group/polyBERT	Academic Research Use License (non-commercial) GitHub	DeBERTa-style fingerprint over PSMILES → multitask head for NSF PAGES >30 properties via Polymer Genome	Per-block T fingerprint

Model	Repo	License	Properties	REWEAR-FUSED role
PolyID (NREL)	github.com/NREL/polyID	BSD-3-Clause (commercially usable)	Multi-output MPNN; 8 properties: Tg, Tm, density, Young's modulus E, P_O2, P_N2, P_CO2, P_H2O; Tg MAE 19.8 °C test / 26.4 °C experimental (nih) (St. John et al., <i>Macromolecules</i> 56:8547-8557, 2023, DOI 10.1021/acs.macromol.3c00994)	PRIMARY block-level predictor – open commercial license
POLYT5	T5 over polymer text	Variable	Generic polymer text → property	Backup
Ding et al. PU-elastomer model	Datasets released as SI to <i>Chin. J. Polym. Sci.</i> 41:422-431, 2023, DOI 10.1007/s10118-022-2838-6	Journal SI; copyright Springer/CCS	Young's modulus, tensile strength, elongation at break for PUEs; (Springer) 5-algorithm ensemble (XGBoost, RF, SVR, ANN, kNN); R ² 0.85-0.91 on benchmark dataset (50.4% of raw retained via Recursive Data Elimination, 20 iterations × 5% removal) (ScienceDirect)	PU-elastomer-specific mechanical predictor – exactly the right tool

Decompose-predict-recompose workflow:

- Decompose** each candidate PU-urea into soft-segment PSMILES (e.g., PCL [\[*OC\(=O\)CCCCO*\]](#)), hard-segment PSMILES (HDI + 1,4-BDO), urea/urethane link.
- Predict** soft-segment Tg via PolyID; hard-segment Tg + modulus via PolyID + Ding XGBoost ensemble.
- Recompose** at the full-chain level: Fox equation for blend Tg of miscible amorphous components; rule-of-mixtures for modulus weighted by hard-segment wt%; Halpin-Tsai for two-phase composites.
- Output** predicted full-fiber soft-Tg, hard-Tg, room-T modulus, elongation, tensile strength.

2(c) Datasets

Dataset	URL	Size	License
PoLyInfo	polymer.nims.go.jp (MatNavi)	~half a million data points, ~100	Free registration

Dataset	URL	Size	License
(NIMS)		properties (Ishii et al., 2024, DOI 10.1080/27660400.2024.2354651)	DICE Account; web scraping prohibited per MIT Terms Nims
PI1M	github.com/RUIMINMA1996/PI1M	~1,000,000 hypothetical polymer PSMILES generated by RNN trained on ~12,000 PoLyInfo polymers ACS Publications	MIT (academic-use only) data per README & Luo, <i>J. Chem. Inf. Model.</i> 60:4684–4694, 2020, DOI 10.1021/acs.jcim.3c00101
PolyID curated dataset	bundled with github.com/NREL/polyID	1,791 unique polymers across 8 properties nih	BSD-3-Clause
Ding et al. PU-elastomer datasets	SI of CJPS 2023 paper	Raw (literature-compiled) + 50.4%-retained benchmark Cjps	Journal SI
Ramprasad group / Polymer Genome	linked from polymergenome.org	~13,766 known polymers Phys.org + 100M BRICS-generated PSMILES	Various

2(d) Enumeration-and-screening workflow

```

For (HDI | H12MDI | IPDI)
  × (PCL-2k | PCD-2k)
  × (1,4-BDO | EG | DAH | DAB)
  × hard_wt% ∈ {25, 27.5, 30, 32.5, 35}:

```

1. Build PSMILES (soft, hard, full repeat) via RDKit
2. Predict Tg_soft, Tg_hard, modulus, elongation via PolyID + Ding XGBoost
3. Compute "carbamate accessibility score":
 - carbonyl-C distance to nearest hard-segment N
 - fraction of carbamates in amorphous (soft) phase
 - SASA of the carbamate C
4. Filter:
 - Tg_soft < -50 °C
 - Tg_hard > 150 °C
 - Elongation > 400%
 - Hard 25-35 wt%
 - Accessibility > threshold τ
5. Rank surviving candidates by carbamate accessibility

Combinatorial size: $3 \times 2 \times 4 \times 5 = 120$ base combinations; with Mn variants and stoichiometric ratios $\rightarrow \sim 10^3$ - 10^4 candidates, all enumerable on CPU in seconds via RDKit (`(rdkit.Chem.AllChem)`).

Cheminformatics stack: RDKit ≥ 2024.09 (Landrum, github.com/rdkit/rdkit), BSD-3-Clause) for SMILES parsing, 3D embedding, Morgan/topological descriptors, SASA. **Pandas** + **scikit-learn** for surrogate ensembling. **XGBoost** ≥ 2.0 for the Ding-style PU-elastomer regressors.

2(e) Engine A $\rightarrow$ Engine B handoff (the co-design coupling)

This is the architectural keystone that turns REWEAR-FUSED from "two separate ML pipelines" into "a single co-design loop."

1. **Engine A's top-ranked fiber** carries a specific carbamate linkage ($(R_1-NH-C(=O)-O-R_2)$, R_1 from HDI, R_2 from PCL).
2. **Engine A emits** `substrate.json` containing:
 - Full PSMILES of the cleavable repeat
 - "Cleavage-target" SMILES of the *minimum hydrolyzable substrate fragment* (model dicarbamate retaining the chemical environment around the bond)
 - 3D coordinates of the substrate fragment with the carbamate carbonyl C marked as the catalytic attack atom
 - R_1 (soft-segment alkyl) and R_2 (hard-segment alkyl/aryl) tagged so the Engine B pocket can be tailored to accommodate them
3. **Engine B reads** `substrate.json` and builds the theozyme: Ser-O γ 3.0 Å from carbamate C (attack geometry), His-N ϵ 2.8 Å from Ser-O γ , Asp-O δ 2.7 Å from His-N δ , two oxyanion-hole

backbone amides 2.9 Å from carbonyl O.

4. **Engine B emits a paired enzyme PDB + JSON** with the carbamate substrate docked.
5. **The matched pair** is what gets validated (Boltz-2 + PLACER + FoldSeek) and shown in the demo.

Engine A's chemistry choice fully determines Engine B's design target — a single co-design loop.

PART 3 — ENGINE C: DIGITAL RECYCLABILITY PASSPORT

3(a) Public data sources with APIs

Source	Endpoint	Format	Notes
CPSC Recalls API	https://www.saferproducts.gov/RestWebServices/Recall?format=json (also ?format=xml)	JSON/XML	Filterable Recall ~9,000+ cpsc.gov/public/
EU Safety Gate (RAPEX)	ec.europa.eu/safety-gate-alerts/screen/search ; weekly Excel; mirror at public.opendatasoft.com/explore/assets/healthref-europe-rapeX-en/	Excel + scraped JSON	~2,000 n from EC;
OEKO- TEX Standard 100	oeko-tex.com (Class I limits catalogue, public PDF)	PDF	Class I (i
Toxic- Free Future/ Peaslee lab	Paper SI tables (e.g., Whitehead et al., <i>Environ. Sci. Technol. Lett.</i> 2021, DOI 10.1021/acs.estlett.0c00876, on PFAS in children's textiles)	CSV (manual)	Labeled 1
ECHA SCIP database	echa.europa.eu/scip-database ; bulk dump at echa.europa.eu/scip-database-data	XLSX/CSV	Article-16

Source	Endpoint	Format	Notes
EU CN HS codes	taxation- customs.ec.europa.eu/taxation/customs/economic- customs-issues/combined-nomenclature_en	XML	Garment

3(b) Regulations to encode (limit values)

Regulation	Scope	Key limits
CPSIA 2008 (US)	Children's products	Pb ≤ 100 ppm in substrate (16 CFR 1303); phthalates DEHP/DBP/BBP/DINP/DIDP/DnHP/DCHP ≤ 0.1% in toys/childcare (16 CFR 1307); 16 CFR 1610 clothing flammability
California AB 1817 (Cal. Health & Safety Code §108970)	New textile articles sold in CA	100 ppm Total Organic Fluorine (TOF) from Jan 1, 2025; 50 ppm from Jan 1, 2027; LegiScan outdoor-severe-wet apparel exempt until Jan 1, 2028. LegiScan Certificate of Compliance required from manufacturer. SGS
California AB 347	PFAS enforcement	DTSC registration by July 1, 2029; enforcement starts July 1, 2030; penalties ≥ \$10,000/violation
New York ECL §37-0121	New apparel sold in NY	Bans intentionally-added PFAS from Jan 1, 2025; Morgan Lewis DEC must set numerical TOF threshold by Jan 1, 2027 Morgan Lewis
Maine LD 1503 (Title 38 §1614)	All consumer products	Mandatory reporting of intentionally-added PFAS; phased bans through 2030
REACH Annex XVII Entry 43 (EU)	Azo dyes in textiles	~22 azo dyes restricted; aromatic-amine release ≤ 30 mg/kg in finished product
REACH Annex XVII Entry 72 (EU)	CMR substances in skin-contact textiles	~33 CMR substances (Cd, Cr(VI), formaldehyde, certain phthalates, certain solvents) limited in clothing/footwear/accessories
OEKO-TEX Standard 100, Class I	Voluntary; de-facto industry standard for infants	Free formaldehyde ≤ 16 ppm; pH 4.0-7.5; tight Ni release; APEO < 100 mg/kg
ESPR Regulation	All products including	Full application July 19, 2026; EU Central DPP Registry go-live same day; textile delegated act targeting 2027 enforcement

Regulation	Scope	Key limits
(EU) 2024/1781	textiles	
EU DPP textile delegated act	Garments sold in EU	Fiber composition ($\geq 1\%$ threshold), Caruma country of origin, carbon footprint (kg CO ₂ e/unit), recycling/end-of-life instructions, repairability/durability index, Caruma REACH-compliant hazardous-substance declarations, GS1 Digital Link Fiengenbaum Solutions or interoperable identifier (QR/NFC/RFID)

3(c) ML approach

Architecture: multi-label gradient-boosted classifier mapping garment record $\rightarrow$ per-regulation risk + overall route $\in \{\text{clear}, \text{lab_test}, \text{divert}\}$.

Inputs: fiber composition vector; country of origin (one-hot from CN HS codes); manufacturer ID + historical recall count (joined from CPSC + RAPEX); product category (children's apparel sub-class: infant onesie, toddler stretchwear, sleepwear, etc.); year; declared chemical/treatment list (DWR, antimicrobial, dye class, flame retardant); test history if available (Pb ppm, TOF ppm, formaldehyde ppm, phthalate %).

Labels (multi-label, one binary per regulation):

- cpsia_pb_risk, cpsia_phthalate_risk, cpsia_flammability_risk
- ab1817_pfas_risk_100ppm, ab1817_pfas_risk_50ppm
- ny_ecl_37_0121_risk
- reach_annex_xvii_43_risk, reach_annex_xvii_72_risk
- oekotex_class1_risk
- Plus secondary action classifier $\rightarrow \{\text{clear}, \text{lab_test}, \text{divert}\}$

Model: LightGBM 4.3+ (preferred over XGBoost for native categorical handling and faster training on tabular data of this size). One-vs-rest multi-output stacking. Severe class imbalance ($< 5\%$ positives per label) handled via scale_pos_weight per-label + SMOTE/ADASYN on the minority labels during cross-validation. Isotonic calibration for the route decision.

Why PR-AUC, not ROC-AUC: positives $< 5\%$ per label; ROC-AUC is misleadingly high for imbalanced binary tasks because true-negative rate dominates. **Precision-Recall AUC** is the standard for needle-in-haystack flagging (Saito & Rehmsmeier, *PLoS ONE* 10:e0118432, 2015, DOI 10.1371/journal.pone.0118432).

Realistic achievable performance on a properly stratified synthetic test set (training on RAPEX + CPSC + AB 1817/CPSIA-aware simulated examples):

- ROC-AUC: 0.85-0.92 per label (inflated by imbalance)
- **PR-AUC: 0.55-0.75 per label** (the honest number to publish)
- For high-stakes labels (ab1817_pfas_risk_50ppm), expect PR-AUC ~0.50 given limited public labeled data — flag honestly in the spec.

3(d) Digital Product Passport schema

Aligned to ESPR textile delegated act draft fields (CIRPASS-2 EU DPP Core Ontology, March 2025) and the EU Central DPP Registry (go-live July 19, 2026). Use **GS1 Digital Link** URI pattern with serialized GTIN.

json


```
{
  "dpp_id": "https://id.gs1.org/01/<GTIN>/21/<SerialNumber>",
  "dpp_version": "ESPR-Textile-2027-v1",
  "product": {
    "gtin": "01234567890123",
    "name": "Carter's Toddler Stretch Onesie – REWEAR-FUSED Edition",
    "category_cn_code": "61112000",
    "manufacturer": {"name": "...", "country": "..."}
  },
  "composition": [
    {"fiber": "cotton", "percent": 85.0},
    {"fiber": "elastane_rewear_fused", "percent": 15.0,
      "psmiles": "...", "engine_a_design_id": "fiber_0042"}
  ],
  "chemicals": {
    "intentional": [...],
    "tested": [
      {"substance": "Total Organic Fluorine", "value_ppm": 12, "method": "EPA 1633"},
      {"substance": "Pb", "value_ppm": 4, "method": "ICP-MS"}
    ]
  },
  "carbon_footprint_kg_co2e": 4.7,
  "circularity": {
    "recyclable": true,
    "recycling_route": "rewear_fused_enzymatic",
    "enzyme_design_id": "urethanase_RF2_0017",
    "expected_recovery_yield_pct": 92
  },
  "compliance": {
    "cpsia": "pass",
    "ab1817_2027": "pass",
    "reach_annex_xvii_43": "pass",
    "reach_annex_xvii_72": "pass",
    "oekotex_class_1": "pass"
  },
  "data_carrier": {"type": "QR", "uri": "<GS1 Digital Link>"},
  "issued_at": "2026-06-17T00:00:00Z",
  "signature": "<JWS over the document>"
}
```

3(e) Backend API: FastAPI architecture

```
rewear-fused-backend/
├─ pyproject.toml
├─ app/
│   ├─ main.py                # FastAPI app
│   └─ api/
│       ├─ engine_a.py        # GET /api/fibers, /api/fibers/{id}
│       ├─ engine_b.py        # GET /api/enzymes, /api/pair/{id}
│       ├─ engine_c.py        # POST /api/classify, GET /api/dpp/{gtin}
│       └─ codesign.py        # GET /api/codesign/{pair_id}
│   └─ models/
│       ├─ compliance_model.pkl # LightGBM
│       └─ polymer_surrogate.pkl # Stacked XGBoost+PolyID
│   └─ artifacts/{fibers,enzymes,pairs,dpps}/*
│   └─ schemas.py             # Pydantic models
```

Stack: Python 3.11 · FastAPI 0.111+ · Uvicorn (dev) + Gunicorn+Uvicorn workers (prod) · Pydantic v2 · RDKit 2024.09 · PyTorch 2.3 with CUDA 12.1 · LightGBM 4.3+ · scikit-learn 1.5+ · pandas 2.2+ · numpy 2.0+ · httpx + `(diskcache)` for CPSC/RAPEX clients · Docker (CPU API image + GPU design image) · optionally Modal/RunPod Serverless for post-demo live design.

Endpoints:

- `(GET /api/fibers/top?n=10)` → list of Engine A candidates with predicted Tg/modulus/elongation
 - `(GET /api/enzymes/top?n=10)` → Engine B designs with pLDDT, PAE, PLACER preorg score, FoldSeek best-hit
 - `(GET /api/codesign/{pair_id})` → matched fiber+enzyme JSON
 - `(POST /api/classify)` (body: garment metadata) → multi-label risk + route decision
 - `(GET /api/dpp/{gtin})` → full ESPR-style DPP JSON
 - `(GET /docs)` → OpenAPI / Swagger auto-generated
-

PART 4 — BACKEND SYSTEM ARCHITECTURE & INTEGRATION

4(a) Overall stack and GPU cloud choice

Component	Choice	Justification
Language	Python 3.11	All ML/compbio libs first-class
Web framework	FastAPI 0.111+	ASGI, Pydantic v2 typed schemas, auto-OpenAPI
Server	Uvicorn (dev) + Gunicorn+Uvicorn workers (prod)	Async, low-overhead
ML/CompBio	PyTorch 2.3 + CUDA 12.1, RDKit, LightGBM, scikit-learn, BioPython, OpenMM (optional MD), Foldseek	All open-source
Containers	Docker; two images: <code>rewear-fused-api</code> (CPU, ~1 GB), <code>rewear-fused-design</code> (CUDA, ~8 GB with RFdiffusion2 + Boltz-2 weights)	Reproducibility, easy deploy to RunPod/Modal
GPU cloud	Primary: RunPod Community Cloud (H100 PCIe spot \$1.99/hr, H100 SXM spot \$2.69/hr, A100 80GB \$1.19-1.89/hr); <code>Hivenet</code> backup: Lambda Labs (H100 80GB \$2.99/hr on-demand, April 2026); <code>SynpixonCloud</code> spot-fallback: Vast.ai (\$0.90-4.10/hr)	Cheapest H100 spot; per-second billing; <code>Flexprice</code> pre-built CUDA templates
Storage	Local FS during build; AWS S3 or RunPod network volume for artifacts	Artifacts <500 MB total
Frontend	Static artifacts via FastAPI <code>StaticFiles</code> or CloudFront; frontend pulls and renders, never triggers GPU work	Eliminates demo-time GPU dependency

Realistic weekend GPU spend: $4 \times \text{H100 SXM} \times 40 \text{ h} \times \$2.69/\text{hr} \approx \$430 + \sim \50

CPU/storage/egress = **$\sim \$500$ total. AlphaFold3 (closed weights) and ESM3 (restricted) are *not used*** to keep the stack commercially redistributable.

4(b) Pre-computed artifacts and data contract

Hard rule: the live demo must NOT depend on a real-time GPU job. Network hiccups, queue waits, or model failures during a 5-minute pitch destroy a hackathon submission. The backend ships precomputed, signed JSON+PDB artifacts.

```

artifacts/
├─ fibers/fiber_{0001..0010}.json      # PSMILES, properties, BigSMILES,
accessibility score
├─ enzymes/urethanase_RF2_{0001..0010}.pdb  # designed structure
├─ enzymes/urethanase_RF2_{0001..0010}.json  # sequence, pLDDT, PAE, PLACER,
FoldSeek, kcat/KM estimate
├─ pairs/pair_{0001..0010}.json          # {fiber_id, enzyme_id,
substrate_smiles, docking, validation}
└─ dpps/dpp_example_*.json

```

Data contract (Pydantic v2, frozen day 1):

python

```

class Fiber(BaseModel):
    fiber_id: str
    psmiles_soft: str; psmiles_hard: str; bigsmiles_full: str
    hard_segment_wt_pct: float
    tg_soft_C: float; tg_hard_C: float
    youngs_modulus_MPa: float; tensile_strength_MPa: float; elongation_at_break_pct:
    carbamate_accessibility_score: float
    cleavage_substrate_smiles: str; cleavage_substrate_pdb_block: str

class Enzyme(BaseModel):
    enzyme_id: str; sequence: str; length: int
    plddt_mean: float; plddt_active_site: float; pae_triad_max: float
    boltz2_confidence: float
    placer_preorg_rmsd: float; placer_oxyanion_hbonds: int
    foldseek_best_tm_score: float; foldseek_best_hit_pdb: str; foldseek_best_hit_seq
    novelty_verdict: str # "novel" if TM < 0.5 and seq_id < 30%

class CoDesignPair(BaseModel):
    pair_id: str; fiber: Fiber; enzyme: Enzyme
    substrate_smiles: str; docking_energy_kcal_mol: float; validation_status: str

```

4(c) Co-design loop orchestration

```

[Engine A – 5-6 h CPU]    →  emits fibers/*.json + substrate.json
    ↓
[Engine B – 30-40 h on 4× H100]
    RFdiffusion2 → LigandMPNN → Boltz-2 → PLACER → FoldSeek
    ↓

```

```

[Pair construction – 1 h CPU]   →  pairs/*.json
      ↓
[Engine C training – 2-3 h CPU] →  LightGBM on CPSC + RAPEX + simulated
      ↓
[DPP generator – minutes CPU]   →  dpps/*.json
      ↓
[FastAPI serves all artifacts; frontend renders]

```

A single Python (`run_pipeline.py`) driven by a YAML config executes each stage as a separate process. State persisted to local disk; checkpoints between stages so any single failure does not lose more than ~2 hours.

4(d) Realistic 72-h backend build sequence and where the risk is

Hour	Task	Risk
0-4	Repo scaffold, Docker images, FastAPI skeleton, frozen Pydantic schemas, frontend handoff	Low
4-10	Engine A: RDKit enumerator + PolyID/polyBERT surrogates + Ding-XGBoost fine-tune on PU-elastomer benchmark	Medium — PolyID install fiddly (TensorFlow 2.14 dependency); fallback to <code>nfp</code> + pretrained weights
10-14	Engine A run; emit top 10 fibers + substrate.json	Low
8-10 (parallel)	RunPod GPU spin-up; pull RFdiffusion2 SIF, LigandMPNN, Boltz-2, PLACER weights; smoke-test Lauko 2025 example	HIGH — SIF download multi-GB; apptainer may need root; Boltz-2 first-run downloads ~3 GB
14-50	Engine B run on 4× H100 (RFdiffusion2 → LigandMPNN → Boltz-2 → PLACER → FoldSeek)	HIGHEST — single-tool failure costs ~6 h; mitigate with checkpoints; halve scale if slow at hour 20
50-55	Engine C: build dataset from CPSC+RAPEX APIs; train LightGBM multi-label; calibrate	Low
55-60	DPP generator; assemble paired artifacts; sign with JWS	Low
60-66	API endpoints; OpenAPI; staging deploy	Low
66-72	End-to-end smoke test with frontend; record fallback static demo; freeze	Medium

Biggest risk is hour 8–50 (Engine B GPU pipeline). Mitigations:

1. **Smoke-test with the Lauko 2025 example first** — confirm RFdiffusion2 + LigandMPNN + Boltz-2 + PLACER all run end-to-end on a known input before launching the REWEAR-FUSED job.
 2. Use (github.com/laukoag/serine-hydrolase-design) as a tested wrapper — already glues these modules together for a serine hydrolase; Engine B is structurally the same problem with a different substrate.
 3. **Two parallel half-scale funnels** (1,000 backbones each on 2× H100) so a crash in one doesn't kill the whole submission.
 4. **Pre-stage docker/apptainer images on the GPU box before the build officially starts** — the RFdiffusion2 SIF is ~10 GB.
-

Recommendations

Stage 1 (Days –7 to 0, pre-hackathon): Set up the GPU box and pre-download all weights. Pull RFdiffusion2 SIF, LigandMPNN weights, Boltz-2 weights, PLACER, FoldSeek databases (AlphaFoldDB v4 + PDB100). Run the Lauko 2025 included example end-to-end on a known input. **Benchmark:** if RFdiffusion2 + LigandMPNN + Boltz-2 + PLACER do not complete on the Lauko sample input in less than 30 minutes on a single H100, escalate before the hackathon starts.

Stage 2 (Hours 0–8): Freeze schemas and emit Engine A results first. The fiber catalogue is the lowest-risk module and the substrate JSON it produces is Engine B's input. Engine C training begins in parallel because its inputs (CPSC + RAPEX) are external.

Stage 3 (Hours 8–50): GPU pipeline. If by hour 20 the throughput is <50% of the budget (fewer than 800 backbones complete), **immediately halve the design count to 1,000 backbones × 4 sequences = 4,000 candidates**. The narrative still works with 5 final designs; do not sacrifice end-to-end completion for raw count.

Stage 4 (Hours 50–72): Engine C, DPP, API, integration. Lock the frontend contract by hour 60; everything after is bug-fixing and the fallback recorded demo.

Benchmarks that would change the plan:

- If RFdiffusion2 cannot be installed at all (apptainer issues), **fall back to RFdiffusionAA with theozyme inverse-rotamer prep**, accepting reduced active-site diversity.

- If Boltz-2 OOMs on the GPU box (24 GB+ VRAM jobs), use Boltz-1 (lower memory) or AF2/ColabFold for self-consistency only and skip co-folding.
 - If PLACER cannot install (Baker-lab repos sometimes have private submodules), substitute Rosetta (`cart_bonded`) + (`coordinate_constraint`) scoring for the preorganization metric and report this honestly in the spec.
-

Caveats — credibly demonstrable in 72 hours vs forward-looking vision

Demonstrably credible in 72 hours (will appear as artifacts in the demo):

- 10 ranked aliphatic PU-urea fiber candidates with PolyID/Ding-XGBoost-predicted Tg, modulus, elongation, and carbamate accessibility scores (Engine A).
- 10 de novo carbamate-hydrolase designs as PDB structures, with Boltz-2 confidence, PLACER preorganization scores, and FoldSeek-confirmed novelty (Engine B).
- A LightGBM multi-label compliance classifier trained on CPSC + RAPEX + simulated data with honest PR-AUC numbers per regulation (Engine C).
- A signed ESPR-aligned Digital Product Passport JSON per garment example, GS1 Digital Link URI, QR-code-renderable.
- A FastAPI service with (`/docs`) auto-OpenAPI, returning artifact JSONs deterministically.

Forward-looking vision (must be labeled as such in the spec and the pitch):

- **Wet-lab validation** of Engine B's designed enzymes (expression in *E. coli*, kcat/KM measurement on the actual carbamate, soak experiments against bulk PU-urea). The literature precedent — Lauko et al., *Science* 2025: kcat/KM up to $2.2 \times 10^5 \text{ M}^{-1} \text{ s}^{-1}$ and kcat = 0.076 s^{-1} at pH 7.0, 25 °C, with "rate enhancement of over 10^8 " for esters; Kim et al., *Nature* 2026: kcat 1.5 s^{-1} , kcat/KM $53,000 \text{ M}^{-1} \text{ s}^{-1}$ for zinc hydrolases — makes the *in silico* claim defensible, but no wet experiment is possible in 72 hours. (`Bakerlab + 2`)
- **Synthesis** of an Engine A candidate fiber and mechanical characterization (Instron tensile testing). The Ding et al. PU-elastomer ML model's R^2 of 0.85–0.91 on the benchmark dataset is the calibration anchor, but the predicted candidates are unsynthesized. (`Springer`)
- **Live MD/QM-MM** of the designed enzyme on the substrate. Possible in principle (OpenMM + ML-MM), but >72 h for meaningful sampling.
- **Real chain-of-custody** integration with retail/recycling partners. The DPP schema aligns with ESPR but the Carter's-specific pilot is a post-hackathon partnership ask.
- **Boltz-2 binding-affinity** to the carbamate is a useful *in-silico* binder/decoy classifier, not a kinetic prediction. We report (`affinity_probability_binary`) and explicitly note that catalytic activity, not binding, is the relevant readout.

Honest model-license disclaimers (must appear in the README and the technical PDF):

- **AlphaFold3 weights: non-commercial only** — *not used*.
- **ESM3 (large): non-commercial** — *not used*.
- **polyBERT: academic-only Research Use License** — used only at the per-block fingerprinting stage; would need replacement or relicensing before any commercial deployment. ([GitHub](#))
- **RFdiffusion2 / RFdiffusionAA**: distributed for research; not currently licensed for commercial reuse — fine for a hackathon submission and academic publication, would require Baker-lab/UW licensing for any commercial product.
- **LigandMPNN, ProteinMPNN, Boltz-1/2, Chai-1, FoldSeek, RDKit, LightGBM, FastAPI, PolyID, PI1M**: all commercially usable (MIT, Apache-2.0, BSD-3-Clause).

This stack is the most ambitious, current, citation-anchored backend the team can plausibly ship in 72 hours. The grand-prize narrative — "**we ran the *Science* 2025 / *Nature* 2026 de novo enzyme design pipeline on a non-natural polyurethane-urea substrate that we ourselves designed, and produced a matched, novel, structurally validated enzyme + a regulatorily-aligned Digital Recyclability Passport**" — is, to the best of current literature evidence, *exactly the right thing to attempt* and exactly the thing that "shouldn't be possible in 72 hours."